

Finding the Derivative at a Point

Directions: Find the slope of the tangent line (or the slope of the function or the instantaneous rate of change of the function) at the given x-value for each equation.

1. $y = 4x^3 - 5x^2 + 2x - 7$ at $x = 2$

2. $y = 3x^{-4} - 2x^{-2} + 5$ at $x = 1$

3. $y = 2x^{\frac{5}{2}} - 3x^{\frac{3}{2}} + x$ at $x = 4$

4. $y = 2\sin x - 3\cos x + \tan x$ at $x = \frac{\pi}{4}$

5. $y = (x^2 + 3x)(x^3 - 1)$ at $x = 1$

6. $y = (2x - 1)\sin x$ at $x = \frac{\pi}{2}$

7. $y = \frac{x^2+1}{3x-1}$ at $x = 1$

8. $y = \frac{\sin x}{x+1}$ at $x = 0$

9. $y = \frac{x+2}{\cos x}$ at $x = 0$

10. $y = \sin(3x)$ at $x = \frac{\pi}{9}$

11. $y = \cos\left(2x - \frac{\pi}{3}\right)$ at $x = \frac{\pi}{3}$

12. $y = \tan(4x)$ at $x = \frac{\pi}{16}$

13. $y = \sec(2x)$ at $x = \frac{\pi}{6}$

14. $y = (3x^2 - 1)^4$ at $x = 1$

15. $y = (2x + 1)^{-3}$ at $x = 1$

16. $y = \cos\left(\frac{\pi x^2}{3}\right)$ at $x = 1$

17. $y = (x^2 + 1)\cos\left(\frac{\pi x}{2}\right)$ at $x = 1$

18. $y = \frac{\sin(2x)}{x^2+1}$ at $x = 0$

19. $y = \frac{x^2+1}{\sin x}$ at $x = \frac{\pi}{2}$

20. $y = (x^2 + 1)^2 \sin(\pi x)$ at $x = \frac{1}{2}$